

① Predictive Parsing

- 1) Eliminate left recursion
- 2) Solve left factoring
- 3) First & Follow function
- 4) Predictive Parsing table
- 5) LL(1) Grammar

First Function

Grammar

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow ID \mid (E)$$

STEP-01:

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid \epsilon$$

$$F \rightarrow ID \mid (E)$$

STEP-03:

$$\text{First}(E) = \text{First}(T) = \text{First}(F) = \{ID, (\}$$

	First Function	Follow
E	{ (, ID }	{ \$,) }
E'	{ +, ϵ }	{ \$,) }
T	{ (, ID }	{ +, \$,) }
T'	{ *, ϵ }	
F	{ (, ID }	

STEP-02:

There is no left factoring

Prayan

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$$P \rightarrow i | o | nTS$$

$$Q \rightarrow p | aS | bScST$$

$$R \rightarrow b | \epsilon$$

$$S \rightarrow c | Rn | \epsilon$$

$$T \rightarrow RSq$$

	First function
P	{i, c, n}
Q	{i, c, n, a, b}
R	{b, \epsilon}
S	{c, b, n, \epsilon}
T	{b, c, n, q}

$$T \rightarrow RSq$$

$$\rightarrow bSq | Sq$$

$$\rightarrow Sq$$

$$\rightarrow cq | bq | nq | q$$

$$S \rightarrow aSe \mid STS$$

$$T \rightarrow RSe \mid Q$$

$$R \rightarrow rSr \mid \epsilon$$

$$Q \rightarrow ST \mid \epsilon$$

STEP-01:

$$S \rightarrow \frac{aSe}{\beta} \mid \frac{STS}{\alpha}$$

$$S' \rightarrow TSS' \mid \epsilon$$

$$T \rightarrow RSe$$

$$\rightarrow rSe \mid Se$$

$$\mid ae$$

$$T \rightarrow Q$$

$$S \rightarrow TSS'$$

$$aSS' \mid rSS' \mid SS'$$

	First function
S	{a}
S'	{a, r, \epsilon}
T	{a, \epsilon, r}
Q	{r, \epsilon}
R	{a, \epsilon}

Rules of Follow Function

- ① First non-terminal (start variable) এর follow function এ "\$" বসে।
- ② উই চিহ্নের ডান দিকের follow বের করতে হবে।
- ③ আর ডান terminal থাকলে সেটাও বসবে।
- ④ ডান কিছু না থাকলে (Null হলে) সে যে rule থেকে আসছে সেই Non-terminal এর follow function বসবে।
- ⑤ যদি ডান Non-terminal থাকলে সেটার ডানের first function বসবে।

follow function এ '\$' হবে না

$$S \rightarrow aSe \mid B$$

$$B \rightarrow bBcf \mid C$$

$$C \rightarrow cCg \mid d \mid \epsilon$$

	first function	follow function
S	{a, B}	{ \$, }
B	{b, c}	
C	{c, d, ϵ }	